PHRÓNĒSIS ACADEMIC HUB

CHM 102: Organic chemistry

1. Define a functional group and explain its importance in organic chemistry.

2. Explain how functional groups determine the physical and chemical properties of organic compounds.

3. Classify organic compounds according to their functional groups, giving two examples for each class.

4. Define alkanes. Discuss their general formula, structural characteristics and sources.

5. Explain the physical properties of alkanes and discuss how molecular size affects these properties.

6. Describe 2 chemical properties of alkanes, illustrating each reaction with balanced equations.

7. Discuss briefly the free-radical mechanism of the halogenation of alkanes.

8. Explain the combustion of alkanes. Differentiate between complete and incomplete combustion, stating their products and environmental implications.

9. Describe the industrial and pharmaceutical importance of alkanes.

10. Explain the cracking of alkanes. State its types and industrial applications.

11. Compare alkanes, alkenes and alkynes with respect to structure, bonding, reactivity and characteristic reactions.

12. Explain why alkanes are generally less reactive than alkenes and alkynes.

13. Discuss the preparation of alkanes in the laboratory and on an industrial scale.

14. Explain the concept of homologous series using alkanes as an example.

15. Discuss the hybridization and geometry of carbon atoms in alkanes.

16. Explain the effect of branching on the physical properties of alkanes.

17. Discuss the sources, composition and fractional distillation of petroleum.

18. Describe the major petroleum fractions obtained during refining and state one important use of each.

19. Explain the relationship between the structure of an organic compound and its functional group.

20. Discuss the importance of functional group interconversion in organic synthesis.

21. With suitable examples, explain how one functional group can be converted into another.

22. Explain the significance of functional groups in the manufacture of pharmaceutical products.

23. Write short notes on:

i. Functional groups

ii. Homologous series

iii. Fractional distillation of petroleum

iv. Free-radical substitution

v. Combustion of alkanes

vi. Cracking of hydrocarbons

vii. Hybridization in alkanes